

Advancing machine learning for protein structure prediction in drug discovery

Specific Aims

Aim 1: Advance machine learning for protein structure prediction in drug discovery via a materials-focused strategy.

Aim 2: Advance machine learning for protein structure prediction in drug discovery via a materials-focused strategy.

Background

This proposal addresses machine learning for protein structure prediction in drug discovery, a problem with substantial significance. We will assemble a multidisciplinary team with complementary expertise. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Prior work has established a partial understanding, yet key gaps remain.

Approach

We will use appropriate methods to achieve the aims.

Innovation

Prior work has established a partial understanding, yet key gaps remain.

Investigator Context

The PI has a sustained track record of funded work in this area. The R2 host institution offers focused departmental resources and regional research partnerships, backed by a growing institutional research base.

Budget Justification

Total requested support is \$998,676 over 4 years at a R2 institution, covering personnel, materials, and dissemination.

References

1. [1] <https://doi.org/10.1109/TPAMI.2016.2577031>
2. [2] <https://doi.org/10.1001/jama.2016.9797>
3. [3] <https://doi.org/10.1101/gr.229102>
4. [4] <https://doi.org/10.1371/journal.pone.0173664>
5. [5] <https://doi.org/10.1038/nature14539>
6. [6] <https://doi.org/10.1101/gr.229102>
7. [7] <https://doi.org/10.1126/science.1259855>

8. [8] <https://doi.org/10.1001/jama.2016.9797>
9. [9] <https://doi.org/10.1038/nature14539>